

System Description/Purpose

We are going to build a system that will act as a data analysis agent. It will appear as a chat system, with file upload and text entry options, with a UI to display the output of the agent.

Users will be able to upload data files and ask a question, then the agent will create Python code to solve the user's problem. The agent will then use a local environment to load and process the data according to the user's prompt, before showing and explaining the results of the code execution to the user.

Source Material

To get a mixture of simple and complex questions, we plan to source our data from several sources. Documentation for various libraries pertaining to data science in Python (such as Pandas and Matplotlib) will be the primary source for simple data science queries. For more complex queries, we will source questions and answers from stack-overflow filtered with tags to ensure we are getting domain-specific questions with answers in Python. In both cases, we will need to convert the data we are scraping into a uniform format of questions in plain text matched with Python code that performs the desired task.

Vision Statement

A successful implementation of the system will allow text entry and file upload such that the user can upload files with data (i.e., CSV, JSON) and ask a question about the data in plain text. The model will take this as its input and generate Python code that answers this query as output. The actual code generated will be hidden from the user and executed under the hood such that the user is only shown the final output of the code that answers their question, which can be in the form of raw numbers, tables, or plots.

Example Input:

- Show the average sales by region.

- [attached file: sales_data.csv]

Example Output:

Region	Sales
North	12,345.6
South	78,901.2
East	34,567.8
West	90,123.4